



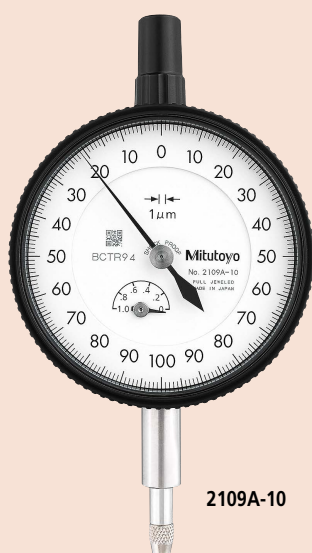
An inspection certificate is supplied as standard.
Refer to page U-9 for details.

Dial Indicators

Comparison measuring instruments which ensure high quality, high accuracy and reliability.

SERIES 2 — Standard Type, 0.001 mm & 0.005 mm Graduation

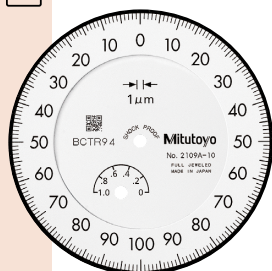
- Standard 0.001 mm and 0.005 mm graduation dial indicators having a bezel with an outside diameter of 57 mm.
- The bezel clamp and lifting lever (optional) can be attached to either the right or left side. These parts can be easily installed and removed without tools.
- Watertight assembly of bezel and crystal as well as the use of an O-ring prevents water or oil penetration.
- The stem and spindle are made of high-strength quench-hardened stainless steel suitable for heavy-duty use.
- A carbide contact point is used.
- A special alloy is used for the sector gears to provide improved wear resistance.
- The indicator uses jeweled bearings, providing excellent indication sensitivity and durability.
- Application of an anti-reflective and hard surface coating improves scale visibility along with scratch and chemical resistance.



2109A-10



Balanced scale

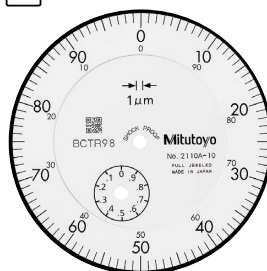


Graduation: 0.001 mm, Measuring range: 1 mm

2109A-10

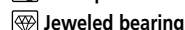
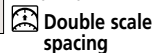


Continuous scale

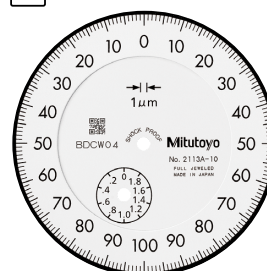


Graduation: 0.001 mm, Measuring range: 1 mm

2110A-10

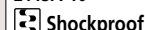


Balanced scale

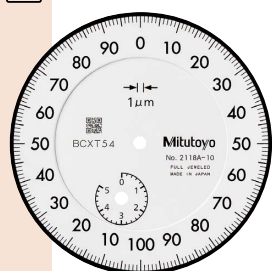


Graduation: 0.001 mm, Measuring range: 2 mm

2113A-10



Continuous scale

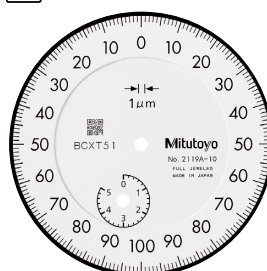


Graduation: 0.001 mm, Measuring range: 5 mm

2118A-10



Balanced scale

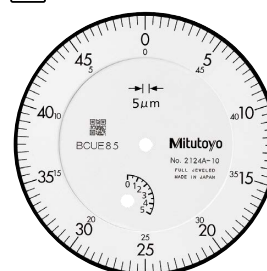


Graduation: 0.001 mm, Measuring range: 5 mm

2119A-10



Continuous scale



Graduation: 0.005 mm, Measuring range: 5 mm

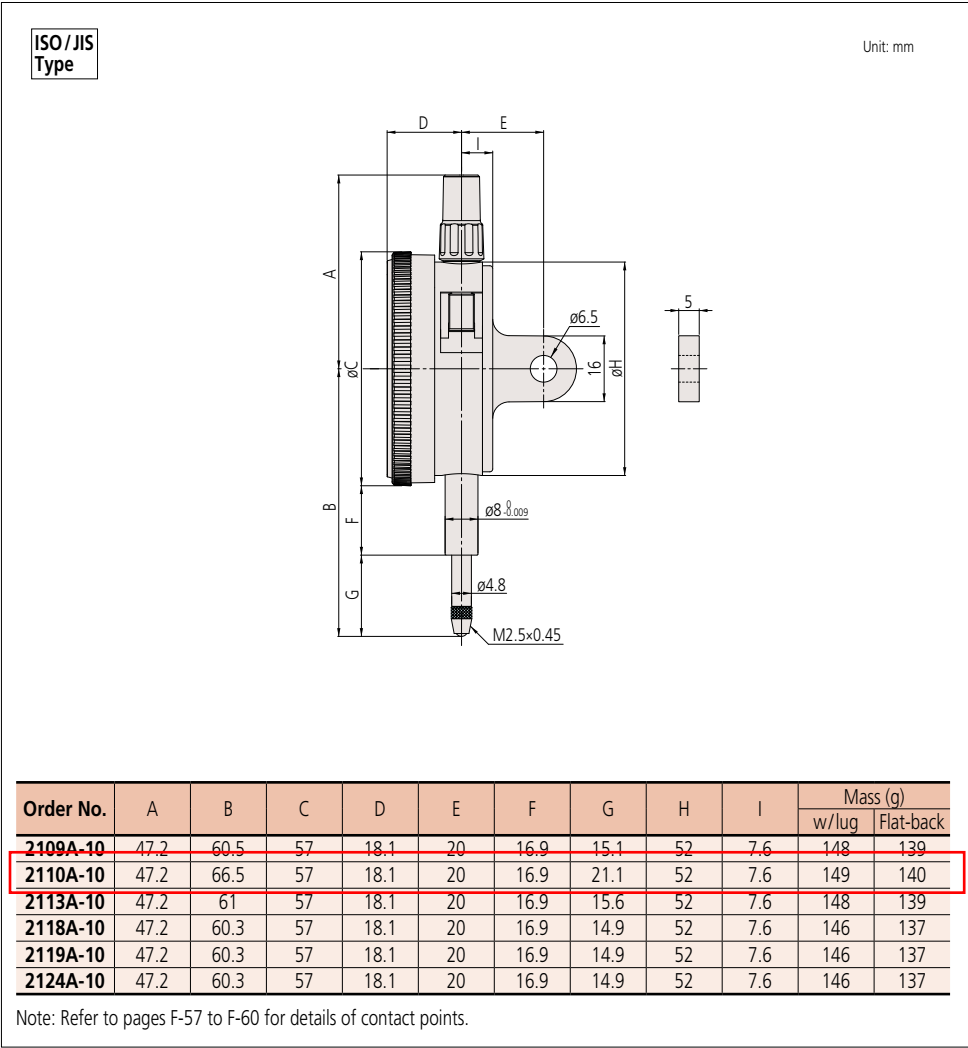
2124A-10



Dial Indicators

Comparison measuring instruments which ensure high quality, high accuracy and reliability.

DIMENSIONS



Optional Accessories

- Limit hand (2 pcs.): 21AZB195
- Bezel clamp: 21AZB148

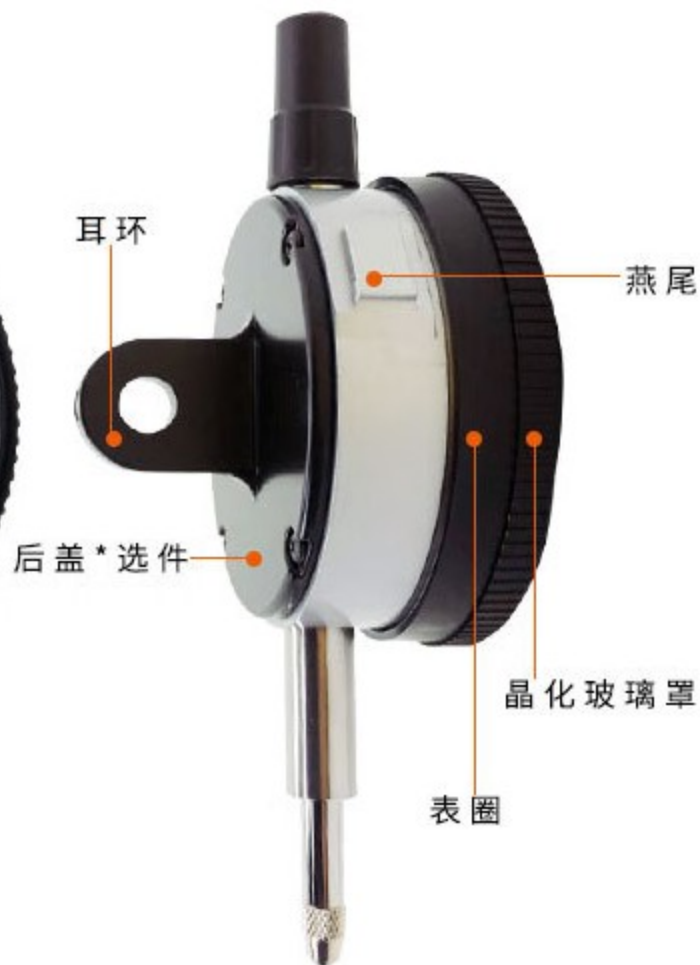
FEATURES

Metric							
Order No.							
w/lug	Flat-back	ISO/JIS type	Flat-back	ISO/JIS type	Flat-back	ISO/JIS type	Flat-back
2109A-10	2109AB-10	✓	✓	✓	✓	✓	✓
2110A-10	2110AB-10	✓	✓	✓	✓	✓	✓
2113A-10	2113AB-10	✓	✓	✓	✓	✓	✓
2118A-10	2118AB-10	✓	✓	✓	✓	✓	✓
2119A-10	2119AB-10	✓	✓	✓	✓	✓	✓
2124A-10	2124AB-10	✓	✓	✓	✓	✓	✓

SPECIFICATIONS

Metric											
Order No.											
		Graduation (mm)	Range (range/rev) (mm)	Maximum permissible error (MPE) (μm)							
				Indication error				Hysteresis	Repeat-ability	Dial reading	Measuring force (N)
				1/10 Rev	1/2 Rev	1 Rev	Measuring range				
2109A-10	2109AB-10	0.001	1 (0.2)	2	3	4	5	2	0.5	0-100-0	1.5 or less
2110A-10	2110AB-10	0.001	1 (0.1)	2	3	4	5	2	0.5	±0-100	1.8 or less
2113A-10	2113AB-10	0.001	2 (0.2)	2	4	5	7	2	0.5	0-100-0	1.5 or less
2118A-10	2118AB-10	0.001	5 (0.2)	3.5	5	6	10	3	1	0-100-100	1.5 or less
2119A-10	2119AB-10	0.001	5 (0.2)	3.5	5	6	10	3	1	0-100-0	1.5 or less
2124A-10	2124AB-10	0.005	5 (0.5)	5	8	9	12	3	3	±0-50	1.5 or less

Note: Completed products inspection is performed in the vertical orientation (contact point downward) and the stated accuracy is guaranteed.



日本三丰mitutoyo

全球知名的综合性量仪制造商

三丰公司成立于1934年，销售遍及世界80个国家和地区，秉承“用精密测量为社会做贡献”的企业经营理念，以“向客户提供最高品质的服务”为宗旨向全球市场提供测量工具以及三坐标测量机、形状测量仪、影像和光学测量仪等丰富的测量设备，设备种类达5500余种。



表蒙多涂层处理



宝石轴承

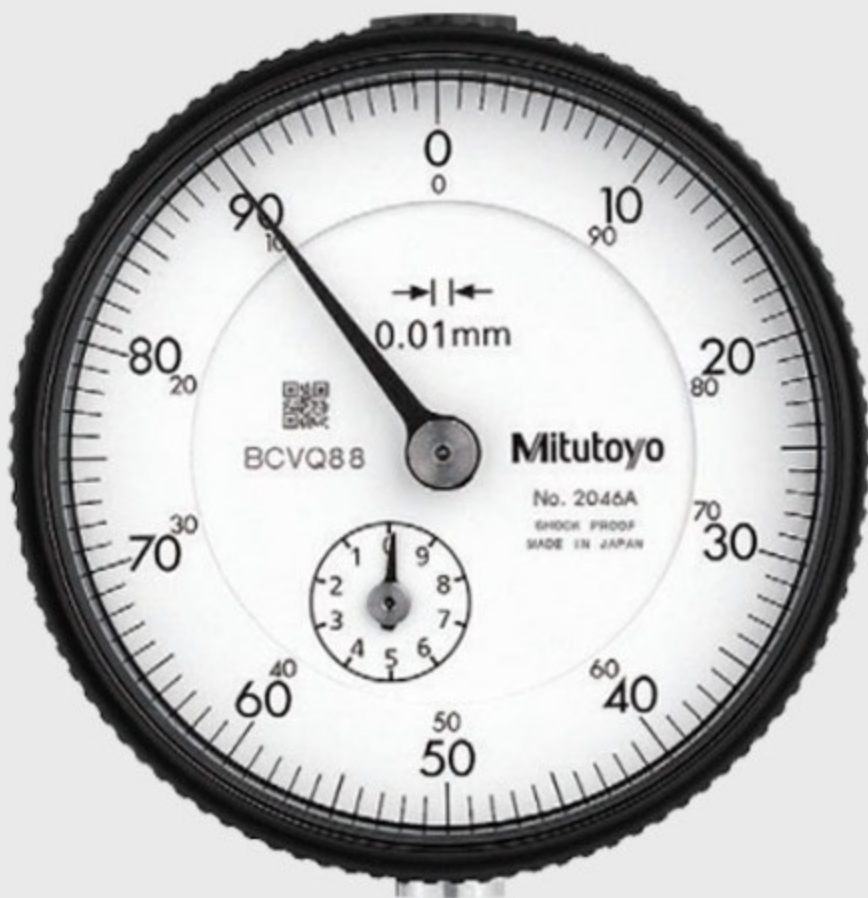


带耳后盖



高强度淬火处理

表蒙多涂层处理



内置宝石轴承

提供高灵敏度指示和耐用性



带耳后盖

可固定指示表，测量更加稳固
(后盖类型可选)



高强度淬火处理

轴套、测杆经过高强度淬火处理
适于高强度测量



产品细节

01

边框表圈



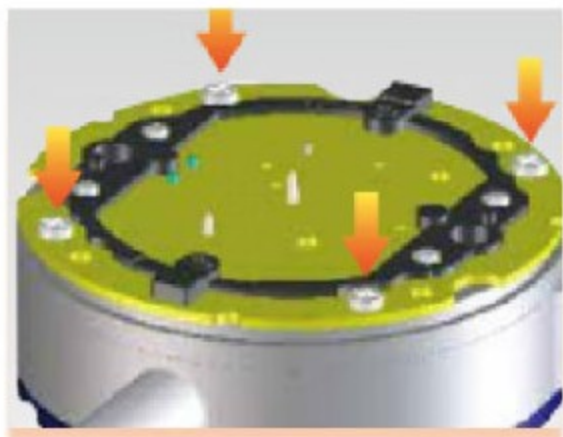
无螺纹通孔，有助于提高防油防尘的能力使用改良的抗冲击性和耐油性的材料，锁紧装置可安在边框左右*表圈锁紧装置为选件

02

革新的轴套衬套设计

更易于轴套夹持，夹持范围更长





03

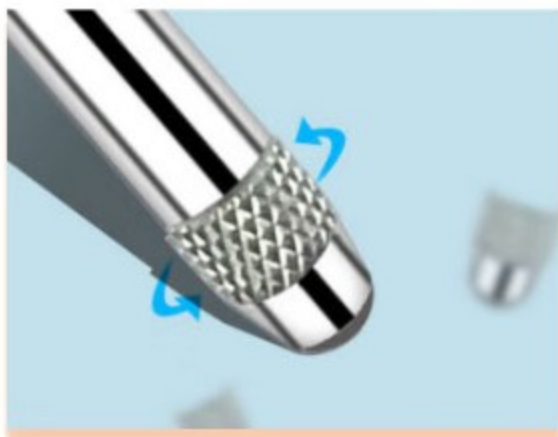
轴承板具

有更高的刚度，可减少回程误差

04

硬质合金测头

不易磨损，还可自行更换测头，
满足不同测量面的需求



05

大表盘

可转动大表盘，手动调零非常便捷



可配置选件多种

后盖	彩色防尘帽	极限标签	测头
	 3971A(B)-2973A(B)不适用		

货号 (带耳后盖)	货号 (平型后盖)	连续刻度	对称刻度	反向读数	单转型	双倍刻度间距型	防震型	防水型(IP6)	防水型(IP64)	防尘型	限位缓冲型	宝石轴承型	峰值保持型	带有同轴旋转计数器	低测力	前置活塞型	指针可调型
2046A	2046AB	✓															
2046A-09	2046AB-09	✓					✓										
2047A	2047AB		✓														
2902A	2902AB			✓													
2310A-10	2310AB-10	✓										✓		✓			
2044A	2044AB	✓															
2044A-09	2044AB-09	✓					✓										
2045A	2045AB		✓														



连续刻度

全量程连续测量型，更易于测量大变化量。该图标正文中省略。



对称刻度

刻度以0为基准左右等分，更方便比较测量



反向读数

适用于孔深或阶差测量，如深度尺一样便利的指示表。



防震型

测杆急剧上压时采用了保护齿轮的缓冲机构，耐冲击性更强。



防水型(IP64)

防止油水各方向的侵入，测杆等可动部位也有密封圈等防护措施。



宝石轴承型 耐久性强



带有同轴旋转计数器
长针和短针同轴旋转，更易读数。



全量程指示误差

13μm
远超出JIS规格的15μm的高精度。

经久耐用

表蒙采用加硬涂层，具有良好的高度防划和耐化学腐蚀性能。

加硬涂层



淬火工艺

轴套、测杆经过高强度淬火处理
硬质合金测针，更耐磨

高强度测量



紧密黏合

表盘和表蒙使用O型圈紧密黏合，可保证表盘面有效防水及防油。

O型圈

